

B.TECH

Regular Theory Examination (Odd Sem-V), 2016-17

Solution of Web Technology Paper (NCS-504)

Q1.

a) How do you call a package?

A **java package** is a group of similar types of classes, interfaces and sub-packages.

Package in java can be categorized in two form, built-in package and user-defined package.

There are many built-in packages such as java, lang, awt, javax, swing, net, io, util, sql etc.

Simple example of java package

```
//save as Simple.java
package mypack;
public class Simple{
public static void main(String args[]){
System.out.println("Welcome to package");
}
}
```

To Compile: javac -d .Simple.java

To Run: java mypack.Simple

b) Mention the usage of private access specifier.

Methods, variables, and constructors that are declared private can only be accessed within the declared class itself.

Private access modifier is the most restrictive access level. Class and interfaces cannot be private.

Variables that are declared private can be accessed outside the class, if public getter methods are present in the class.

Using the private modifier is the main way that an object encapsulates itself and hides data from the outside world.

c) What is HTML?

HTML is the standard markup language for creating Web pages.

- HTML stands for Hyper Text Markup Language
- HTML describes the structure of Web pages using markup
- HTML elements are the building blocks of HTML pages
- HTML elements are represented by tags
- HTML tags label pieces of content such as "heading", "paragraph", "table", and so on

- Browsers do not display the HTML tags, but use them to render the content of the page

d) What is selector? Mention its types.

In CSS, selectors are patterns used to select the element(s) we want to style.

Selector	Example	Example description
<i>.class</i>	.intro	Selects all elements with class="intro"
<i>#id</i>	#firstname	Selects the element with id="firstname"
<i>*</i>	*	Selects all elements
<i>element</i>	p	Selects all <p> elements
<i>element,element</i>	div, p	Selects all <div> elements and all <p> elements
<i>element element</i>	div p	Selects all <p> elements inside <div> elements
<i>element>element</i>	div > p	Selects all <p> elements where the parent is a <div> element
<i>element+element</i>	div + p	Selects all <p> elements that are placed immediately after <div> elements
<i>element1~element2</i>	p ~ ul	Selects every element that are preceded by a <p> element
<i>[attribute]</i>	[target]	Selects all elements with a target attribute

e) How a java script works?

JavaScript is called a Client-side Scripting Language. That means that it is a computer programming language that runs inside an Internet browser (a browser is also known as a Web client because it connects to a Web server to download pages).

Inside a normal Web page we place some JavaScript code. When the browser loads the page, the browser has a built-in interpreter that reads the JavaScript code it finds in the page and runs it.

f) Differentiate Static web pages and Dynamic Web pages.

Static Web Pages

Static Web pages display the exact same information whenever anyone visits it. Static Web pages do not have to be simple plain text. They can feature detailed multimedia design and even videos.

However, every visitor to that page will be greeted by the exact same text, multimedia design or video

every time he visits the page until you alter that page's source code.

Dynamic Web Pages

Dynamic Web pages are capable of producing different content for different visitors from the same source code file. The website can display different content based on what operating system or browser the visitor is using, whether she is using a PC or a mobile device, or even the source that referred the visitor. A dynamic Web page is not necessarily better than a static Web page. The two simply serve different purposes.

g) What is Java Server Pages Technology?

JavaServer Pages (JSP) is a server-side programming technology that enables the creation of dynamic, platform-independent method for building Web-based applications. JSP have access to the entire family of Java APIs, including the JDBC API to access enterprise databases.

h) Compare Servlet with CGI.

Servlet	CGI (Common Gateway Interface)
Servlets are portable	CGI is not portable.
In Servlets each request is handled by lightweight Java Thread	IN CGI each request is handled by heavy weight OS process
In Servlets, Data sharing is possible	In CGI, data sharing is not available.
Servlets can link directly to the Web server	CGI cannot directly link to Web server.
Session tracking and caching of previous computations can be performed	Session tracking and caching of previous computations cannot be performed
Automatic parsing and decoding of HTML form data can be performed.	Automatic parsing and decoding of HTML form data cannot be performed.
Servlets can read and Set HTTP Headers	CGI cannot read and Set HTTP Headers
Servlets is inexpensive than CGI	CGI is more expensive than Servlets

I) What is the usage of CSS?

CSS is used to define styles for documents, including the design, layout and variations in display for different devices and screen sizes. We can place CSS in the <head> of a document with an embedded style sheet, or attach a separate file that defines the styles with an external style sheet. To link an external style sheet to the document, we simply add a link to the style sheet in the <head> of the document.

An external style sheet has many advantages. Keeping the styles separate from your HTML content:

- Helps avoid duplication
- Makes maintenance easier
- Allows you to make a site-wide change in one place

j) Define ODBC.

Open Database Connectivity (ODBC) is a standard application programming interface (API) for accessing database management systems (DBMS). The designers of ODBC aimed to make it independent of database systems and operating systems. An application written using ODBC can be ported to other platforms, both on the client and server side, with few changes to the data access code.

SECTION B

a) Briefly explain the Internet concepts

The **Internet** is the global system of interconnected computer networks that use the Internet protocol suite (TCP/IP) to link devices worldwide. It is a *network of networks* that consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies. The Internet carries an extensive range of information resources and services, such as the inter-linked hypertext documents and applications of the World Wide Web (WWW), electronic mail, telephony, and peer-to-peer networks for file sharing.

What is Internet?

The Internet is essentially a global network of computing resources. You can think of the Internet as a physical collection of routers and circuits as a set of shared resources.

Some common definitions given in the past include –

- A network of networks based on the TCP/IP communications protocol.
- A community of people who use and develop those networks.
- A community of people who use and develop those networks.

Internet-Based Services

Some of the basic services available to Internet users are –

- **Email** – A fast, easy, and inexpensive way to communicate with other Internet users around the world.
- **Telnet** – Allows a user to log into a remote computer as though it were a local system.
- **FTP** – Allows a user to transfer virtually every kind of file that can be stored on a computer from one Internet-connected computer to another.

- **UseNet news** – A distributed bulletin board that offers a combination news and discussion service on thousands of topics.
- **World Wide Web (WWW)** – A hypertext interface to Internet information resources.

WWW stands for **World Wide Web**. A technical definition of the World Wide Web is – All the resources and users on the Internet that are using the Hypertext Transfer Protocol (HTTP).

A broader definition comes from the organization that Web inventor Tim Berners-Lee helped found, the World Wide Web Consortium (W3C): The World Wide Web is the universe of network-accessible information, an embodiment of human knowledge.

In simple terms, The World Wide Web is a way of exchanging information between computers on the Internet, tying them together into a vast collection of interactive multimedia resources.

HTTP stands for **H**ypertext **T**ransfer **P**rotocol. This is the protocol being used to transfer hypertext documents that makes the World Wide Web possible.

A standard web address such as Yahoo.com is called a URL and here the prefix **http** indicates its protocol

URL stands for **U**niform **R**esource **L**ocator, and is used to specify addresses on the World Wide Web. A URL is the fundamental network identification for any resource connected to the web (e.g., hypertext pages, images, and sound files).

A URL will have the following format –

protocol://hostname/other_information

The protocol specifies how information is transferred from a link. The protocol used for web resources is HyperText Transfer Protocol (HTTP). Other protocols compatible with most web browsers include FTP, telnet, newsgroups, and Gopher.

The protocol is followed by a colon, two slashes, and then the domain name. The domain name is the computer on which the resource is located.

Links to particular files or subdirectories may be further specified after the domain name. The directory names are separated by single forward slashes.

Each page available on the website is called a *web page* and first page of any website is called *home page* for that site.

Every Website sits on a computer known as a Web server. This server is always connected to the internet. Every Web server that is connected to the Internet is given a unique address made up of a series of four numbers between 0 and 256 separated by periods. For example, 68.178.157.132 or 68.122.35.127.

When you register a Web address, also known as a domain name, such as tutorialspoint.com you have

to specify the IP address of the Web server that will host the site.

We will see different type of Web servers in a separate chapter.

Web Browsers are software installed on your PC. To access the Web you need a web browsers, such as Netscape Navigator, Microsoft Internet Explorer or Mozilla Firefox.

Currently you must be using any sort of Web browser while you are navigating through my site tutorialspoint.com. On the Web, when you navigate through pages of information this is commonly known as *browsing or surfing*.

SMTP stands for **S**imple **M**ail **T**ransfer **P**rotocol Server. This server takes care of delivering emails from one server to another server. When you send an email to an email address, it is delivered to its recipient by a SMTP Server.

ISP stands for **I**nternet **S**ervice **P**rovider. They are the companies who provide you service in terms of internet connection to connect to the internet.

You will buy space on a Web Server from any Internet Service Provider. This space will be used to host your Website.

HTML stands for **H**yper **T**ext **M**arkup **L**anguage. This is the language in which we write web pages for any Website. Even the page you are reading right now is written in HTML.

This is a subset of Standard Generalized Mark-Up Language (SGML) for electronic publishing, the specific standard used for the World Wide Web.

A hyperlink or simply a link is a selectable element in an electronic document that serves as an access point to other electronic resources. Typically, you click the hyperlink to access the linked resource. Familiar hyperlinks include buttons, icons, image maps, and clickable text links.

DNS stands for **D**omain **N**ame **S**ystem. When someone types in your domain name, www.example.com, your browser will ask the Domain Name System to find the IP that hosts your site. When you register your domain name, your IP address should be put in a DNS along with your domain name. Without doing it your domain name will not be functioning properly.

W3C stands for **W**orld **W**ide **W**eb Consortium which is an international consortium of companies involved with the Internet and the Web.

The W3C was founded in 1994 by Tim Berners-Lee, the original architect of the World Wide Web. The organization's purpose is to develop open standards so that the Web evolves in a single direction rather than being splintered among competing factions. The W3C is the chief standards body for HTTP and HTML.

Benefits of the Internet

1. Helps in reducing the costs of communication, as one can very easily be in touch and communicate

with each other with the help of chatting, electronic mail etc.

2. Acts as a very strong source or a tool at the corporate level, as it helps, inter alia, brings business and helps in generating revenue.
3. Communication and also the coordinating ability between the individuals have been enhanced.
4. Customer satisfaction and customer service have been improved at a very large scale.
5. One can very easily go through or access the various data or the information needed by an individual, hence having a very positive effect in speeding up the whole process of the distribution of knowledge.

Disadvantages of the Internet

1. Legal problems.
2. Problems related to security.
3. Technological problems.
4. Lacking behind in standards as a result of following the same traditional internet culture.

b) Compare and contrast java and javascript

The JavaScript programming language, developed by Netscape, Inc., is not part of the Java platform.

JavaScript **does not create applets or stand-alone applications**. In its most common form, JavaScript **resides inside HTML documents**, and can provide levels of **interactivity to web pages** that are not achievable with simple HTML.

Key differences between Java and JavaScript:

- Java is an OOP **programming language** while Java Script is an OOP **scripting language**.
- Java creates applications that **run in a virtual machine** or browser while JavaScript code is **run on a browser only**.
- Java code needs to be **compiled** while JavaScript code are **all in text**.
- They require different plug-ins.

	Java	JavaScript
Designed by	James Gosling and Sun Microsystems	Brendan Eich
Major Implementations	OpenJDK, many others	Netscape
Typing Discipline	Static, strong, safe, nominative, manifest	dynamic, weak, duck
Appeared in	1995	1995; 18 years ago
Influenced by	Ada 83, C++, C#, Eiffel, Generic Java, Mesa, Modula-3, Oberon, Objective-C, UCSD Pascal, Smalltalk	C, Java, Perl, Python, Scheme, Self
Influenced	Ada 2005, BeanShell, C#, Clojure, D, ECMAScript, Groovy, J#, Javascript, PHP, Python, Scala, Seed7, Vala	ActionScript, CoffeeScript, Dart, JScript .NET, Objective-J, QML, TIScript, TypeScript

code	compiled to bytecode	Codes are all in text, does not need to be compiled
Statement terminators	;	Semicolon terminated
Runs on	Virtual machine	web-browser
Stable release	Java Standard Edition 8 (1.8) / March 18, 2014	1.8.5 (March 22, 2011; 2 years ago)
Introduction (from Wikipedia)	Java is a computer programming language that is concurrent, class-based, object-oriented, and specifically designed to have as few implementation dependencies as possible.	JavaScript is an interpreted computer programming language.

c) **Design a HTML form for a railway reservation system**

<HTML>

<BODY>

<FORM>

<H1>RESERVATION FORM</H1>

<P>NAME<INPUT TYPE="TEXT" SIZE="20" MAXLENGTH="99" />

<P>AGE<INPUT TYPE="TEXT" SIZE="12" MAXLENGTH="99" />

<P>SEX

<SELECT NAME="sex" >

<OPTION>MALE</OPTION>

<OPTION>FEMALE</OPTION></SELECT>

<P>ADDRESS<INPUT

<TYPE="TEXT" SIZE="42" MAXLENGTH="99" />

<P>EMAIL<INPUT TYPE="TEXT" SIZE="30" MAXLENGTH="99" />

```

<P>TELEPHONE<INPUT TYPE="TEXT" SIZE="10" MAXLENGTH="99" />

<P>SELECT YOUR BERTH

<SELECT NAME="CHOICES" SIZE="3">

<OPTION>TATKAL</OPTION>

<OPTION>LADIES</OPTION>

<OPTION>GENRAL</OPTION>

</SELECT></P>

<P>CITY<SELECT naME="CITY"><OPTION SELECTED>

DELHI</OPTION><OPTION>MUMBAI</OPTION><OPTION>KOLKATA</OPTION><OPTION
>CHENNAI</OPTION></p><br><br>

<input type="submit" value="click here to submit" />

<input type="reset" value="clear this form" />

</FORM>

</BODY></HTML>

```

d) Explain in detail the Tomcat Server

Apache Tomcat, often referred to as **Tomcat Server**, is an open-source Java Servlet Container developed by the Apache Software Foundation (ASF). Tomcat implements several Java EE specifications including Java Servlet, JavaServer Pages (JSP), Java EL, and WebSocket, and provides a "pure Java" HTTP web server environment in which Java code can run.

Tomcat is developed and maintained by an open community of developers under the auspices of the Apache Software Foundation, released under the Apache License 2.0 license, and is open-source software. **Apache Tomcat**, is an open source web server and servlet container developed by the Apache SoftwareFoundation. Basically, it implements the Java Servlet and the JavaServer Pages (JSP) specifications from Sun Microsystem, and provides a "pure Java" HTTP web server environment for Java code to run in. In the simplest configuration Tomcat runs in a single operating system process. The process runs a Java virtual machine (JVM) and every single HTTP request from a browser to Tomcat is processed in the Tomcat process in a separate thread. Apache Tomcat includes tools for configuration and management, but can also be configured by

editing

XML

configuration

files.

In simple words, Tomcat acts or behaves as a *development* server on your desktop to use for testing when building applications that use JSF 2, servlets/JSP, or other Java-based dynamic Web technologies. Regardless of what deployment server you use, you'll want a standalone server on your desktop to use for development. By far the best way to use Tomcat for development purposes is from inside Eclipse or another IDE.

e) What do you understand by ASP? What are its advantages and disadvantages?

Active Server Pages

Active Server Pages or ASP.NET is a new programming framework from Microsoft for deploying next generation web applications. It is a structure built on the Common Language Runtime and introduces a new model to server-side web development. It is used with IIS (Internet Information Server) which runs on Microsoft servers.

Advantages of ASP

It provides 3-tiers architecture.

Provide easy way to deploy the new applications and maintenance.

Compatible with around 55 languages.

Consistent Programming Model.

Direct Support for Security.

Rapid development to web applications.

Easy to learn and install.

Disadvantages of ASP

Cost Factor

Running ASP programs costs higher as it requires IIS installed on a Windows platform server which is not free. Moreover, the connectivity of the database is very costly in ASP as you have to buy MS-SQL which is a Microsoft product.

Speed

When it comes to speed, ASP is relatively slower than PHP. This is because ASP can't run fast as it is built on COM based architecture which is an overhead for the server.

Platform Compatibility

ASP is not platform compatible like PHP as and is primarily related with Windows platform with ASP-Apache installed on the server.

Additional Cost

Unlike PHP ASP doesn't have any built-in features such as FTP, email from a webpage and encryption mechanisms and hence, some additional components are required which results in additional expenditure as well.

Base Language

ASP has lots of syntax that are based on Visual Basic which most programmers are not conversant with. Moreover, it is again a product of Microsoft. Probably, this is why many coders prefer PHP over ASP.

ASP has always been a prime choice for e-commerce application development. The controversy as to which one is the better between ASP and PHP still continues. However, both have their respective merits and demerits. Either of these programming languages supports a specific platform. Hence, it is really difficult to say that one is better than the other.

- **Interpreted and Loosely-Typed Code**

ASP scripting code is usually written in languages such as JScript or VBScript. The script-execution engine that Active Server Pages relies on interprets code line by line, every time the page is called. In addition, although variables are supported, they are all loosely typed as variants and bound to particular types only when the code is run. Both these factors impede performance, and late binding of types makes it harder to catch errors when you are writing code.

- **Mixes layout (HTML) and logic (scripting code)**

ASP files frequently combine script code with HTML. This results in ASP scripts that are lengthy, difficult to read, and switch frequently between code and HTML. The interspersing of HTML with ASP code is particularly problematic for larger web applications, where content must be kept separate from business logic.

- **Limited Development and Debugging Tools**

Microsoft Visual InterDev, Macromedia Visual UltraDev, and other tools have attempted to increase the productivity of ASP programmers by providing graphical development environments. However, these tools never achieved the ease of use or the level of acceptance achieved by Microsoft Windows application development tools, such as Visual Basic or Microsoft Access. ASP developers still rely heavily or exclusively on Notepad.

Debugging is an unavoidable part of any software development process, and the debugging tools for ASP have been minimal. Most ASP programmers resort to embedding temporary Response.Write statements in their code to trace the progress of its execution.

- **No real state management**

Session state is only maintained if the client browser supports cookies. Session state information can

only be held by using the ASP Session object. And you have to implement additional code if you, for example, want to identify a user.

- **Update files only when server is down**

If your Web application makes use of components, copying new files to your application should only be done when the Web server is stopped. Otherwise it is like pulling the rug from under your application's feet, because the components may be in use (and locked) and must be registered.

- **Obscure Configuration Settings**

The configuration information for an ASP web application (such as session state and server timeouts) is stored in the IIS metabase. Because the metabase is stored in a proprietary format, it can only be modified on the server machine with utilities such as the Internet Service Manager. With limited support for programmatically manipulating or extracting these settings, it is often an arduous task to port an ASP application from one server to another.

f) Explain COM and DCOM in detail

COM: Component object model is an interface standard for software component introduced by Microsoft in 1993. It is used to enable interprocess communication and dynamic object creation in any programming language that supports the technology. The term COM is often used in the software development industry as an umbrella term that encompasses the OLE, OLE Automation, ActiveX, COM+ and DCOM technologies.

COM is a software architecture that allows applications and systems to be built from the components supplied by different software vendors. It is a set of binary and network standards that allows any software to communicate with each other regardless of the hardware, the operating system, and programming language used for development. COM is not a Programming language but a specification that defines how components can communicate with each other.

In a component based system, components interact with each other by calling methods and passing data. COM ensures that there is a standard method of interaction between the components. All the COM objects need to follow these standards when providing the functionality. Standards are as important to component architecture as they are to any system with interchangeable parts. For example, the signal that a television or radio receives is specified in a standards. Without standards, nothing will function together.

Benefits of COM:

Any Programming language that can create structures of pointers and call functions through pointers can create and use components objects. Components can be implemented in a number of programming languages and can be used for client software written in different programming languages. **This is because COM** provides a binary and not source code standard. This is the fundamental benefit of COM over object oriented language.

DOCM: DCOM is a model as COM but is specially designed for distributed applications. While there is a need of using the applications as distributed then DCOM becomes in picture.

Distributed Component Object Model is proprietary Microsoft technology for communication among software components distributed across networked computers. DCOM, which originally was called “Network OLE”, extends Microsoft’s COM, and provides the communications substrate under Microsoft COM+ application sever infrastructure. It has been deprecated in favour of Microsoft .Net.

In terms of the extensions it added to COM, DCOM had to solve the problem of

- 1. Marshalling:** serialize and de-serializing the arguments and returns values of methods calls over the wire.
- 2. Distributed garbage collection:** ensuring that references held by clients of interfaces are released when, for example, the client process crashed, or the network connection was lost.

g) Describe the PHP life cycle and how does PHP score over ASP and JSP?

Life Cycle of PHP

1. We never start any PHP daemon or anything by ourself. When we start Apache, it starts the PHP interpreter along itself
2. PHP is linked to Apache (In general term SAPI i.e. a Server API) using mod_php5.so module
3. PHP as a whole consists of 3 modules (Core PHP, Zend Engine and Extension Layer)
4. Core PHP is the module which handles the requests, file streams, error handling and other such operations
5. Zend Engine(ZE) is the one which converts human readable code into machine understandable tokens/op-codes. Then it executes this generate code into a Virtual Machine.
6. Extensions are a bunch of functions, classes, streams made available to the PHP scripts, which can be used to perform certain tasks. For example, we need mysql extension to connect to MySQL database using PHP.
7. While Zend Engine executes the generated code, the script might require access to a few extensions. Then ZE passes the control to the extension module/layer which transfer back the control to ZE after completion of tasks.

8. Finally Zend Engine returns back the result to PHP Core, which gives that to SAPI layer, and finally which displays it on your browser.

Score PHP :

PHP

PHP uses a blend of interpretation and compilation in order to provide the best mix of performance and flexibility to programmers.

Behind the scenes, PHP compiles your script down to a series of instructions (called opcodes) whenever it is accessed. These instructions are then executed one by one until the script terminates. This is different from conventional compiled languages such as C++ where the code is compiled down to native executable code then that executable is run from then on. Instead, PHP re-compiles your script each time it is requested.

This constant recompilation may seem a waste of processor time, but it is actually not all that bad because you no longer need to worry about hand recompiling your scripts when you make any changes. On the flip side, many scripts take longer to compile than they do to execute!

Furthermore, it provides very quick feedback during development. If you have an error somewhere in your file, PHP will refuse to compile the page until you have fixed the problem, and you are able to step through execution of your code line by line until you find the problem.

The speed hit of regular compilation is nullified entirely by the use of PHP accelerators.

One major advantage to having interpreted code is that all memory used by the script is managed by PHP, and the language automatically cleans up after every script has finished. This means that you do not need to worry about closing database links, freeing memory assigned to images, and so on, because PHP will do it for you. That is not to say you should be lazy and make PHP do all the work – good programmers clean up themselves, and let PHP work as backup in case something is missed.

- ASP only runs on IIS installed on a Windows platform server
- PHP code runs faster than ASP.
- ASP is built on COM based architecture, which is an overhead for the server whereas PHP code runs in its own memory space.
- PHP programs can run on various platforms like Linux, Unix, Windows and Solaris whereas ASP is mainly associated with Windows platforms.
- PHP has inbuilt features like ftp, email from a web page or even encryption mechanisms but in ASP such features are not built in.
- PHP is based on C++ language and the syntax used in PHP is quite similar to C/C++. ASP's syntax is based on the syntax of Visual Basic.
- PHP can connect to various databases, mainly MySQL, whereas ASP mainly uses MS-SQL.
- PHP codes tend to be easier than ASP codes.

PHP and JSP

PHP can be expanded as **Hypertext Preprocessor**. PHP is a scripting language that is extensively used in the area of Web development. It allows the web developers to quickly write dynamically generated pages.

JSP is JavaServer Pages. It is a technological innovation from JAVA that enables Web developers and designers to develop and maintain, dynamic Web pages that are rich in information which in turn, empower systems.

PHP	JSP
It is the most widely used Web development platform in the world. The barrier to entry is really low.	It is used in broader areas and barrier is higher than PHP.
For this need Server as (typically Apache).	For this need server as Apache Tomcat or JBoss or WebSphere, WebLogic etc.
Immediate feedback and greater productivity as coding less and fast development and execution.	As more coding than PHP need little more time than PHP.
PHP hosting is extraordinarily cheap.	JSP hosting is not much costly than PHP.
Statically typed. Now I don't consider this a \"win\" against dynamic typing, they're just different. But the fact is that if you mistype a variable name in Java it'll not throw any error and will generate a new variable.	Statically typed. Now I don't consider this a \"win\" against dynamic typing, they're just different. But the fact is that if you mistype a variable name in Java it'll generate a compiler error.
Less libraries compare to JSP.	There are a huge amount of third party libraries for Java Web development out there.
Less robust compare to JSP.	IMHO the modern Java Web development approach using, say, Spring (or other IoC container) is extremely robust.
Multithreaded and background processing are somewhat kludgy in PHP.	Multithreaded and background processing are far more natural in Java.
PHP does not allow you to define custom tags.	JSP allow you to define custom tags that can be quite powerful.

Here is a list of reasons why JSP is better than PHP:

- Anything you can do with PHP, you can do with JSP; the reverse is not true
- JSP is much more powerful, since it has access to all the Java libraries. PHP only has access to PHP libraries
- JSP is Object-Oriented, so leads to cleaner code that's easier to debug, maintain, and improve. (PHP also allows objects, but the object model is more primitive, and most scripted pages ignore PHP objects and just use normal variables.)
- The equivalent syntax in JSP is just as simple to learn, so you can get up and running just as quickly -- that is, there's no **extra** startup cost in using Java, at least not a significant one
- Java programmers (as opposed to 15-year-old hackers or HTML monkeys) appreciate the importance of a clean language with complex OO data structures and strong typing